Benchmarking Cognitive Biases in Large Language Models as Evaluators

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Abstract

Large Language Models (LLMs) have recently been shown to be effective as automatic evaluators with simple prompting and in-context learning. In this work, we assemble 16 LLMs encompassing four different size ranges and evaluate their output responses by preference ranking from the other LLMs as evaluators, such as *System Star is better than System Square*. We then evaluate the quality of ranking outputs introducing the COgnitive Bias Benchmark for LLMs as EvaluatoRs (COBBLER)[1], a benchmark to measure six different cognitive biases in LLM evaluation outputs, such as the EGOCENTRIC bias where a model prefers to rank its own outputs highly in evaluation. We find that LLMs are biased text quality evaluators, exhibiting strong indications on our bias benchmark (≈ **40%** of comparisons made by all models) within each of their evaluations that question their robustness as evaluators. Furthermore, we examine the correlation between human and machine preferences and calculate the average Rank-Biased Overlap (RBO) score to be 44%, indicating that machine preferences are misaligned with humans. According to our findings, LLMs may still be unable to be utilized for automatic annotation aligned with human preferences.

1 Introduction

Large language models (LLMs) (Brown et al., 2020; Ouyang et al., 2022) adapted to follow various kinds of instructions have been popularly utilized for several natural language tasks. The general standard for testing a model's capabilities is benchmarking its performance on static evaluation suites such as Fan et al. (2019) and Wang et al. (2020). With the increased usage of language models as general-purpose assistants and their artificial nature (Das et al., 2024), current task-specific benchmarks are insufficient to measure the quality of generated texts in the wild.

Recent studies have shown that LLMs can serve as evaluators themselves: Wu and Aji (2023) utilize LLMs as self-evaluators to automatically judge the quality of open-ended generations and compare them with human judgments via an Elo-score calculation. Other works, such as AlpacaEval (Li et al., 2023b), also utilize LLMs, such as GPT-4 (OpenAI, 2023), as automatic evaluators to reduce the time and cost overhead of human annotations. As noted by these works, such automatic evaluation leaderboards have a number of limitations, including a preference for long outputs or outputs that are more similar to the evaluators' generation qualities.

In this work, we propose COBBLER, the COgnitive Bias Benchmark for evaluating the quality and reliability of LLMs as EvaluatoRs, as depicted in Figure 1. We collect a set of 50 question-answering instructions from two well-established benchmarking datasets: BIGBENCH (Srivastava et al., 2023) and ELI5 (Fan et al., 2019). We then generate responses from 16 open- and closed-source LLMs and conduct a round-robin over every possible unique pair between each of the model responses, prompting each model to evaluate its own and other models' responses.

We then test six different biases to benchmark their evaluation quality and categorize the model biases into two groups: (1) **Implicit Biases**, which can be implicitly extracted from each model's evaluation via a vanilla prompt, and (2) **Induced Biases**, which add modifications to the original prompts akin to induce negative behaviors. As shown in Figure 2, we find that the majority of the models strongly exhibit several of the different biases, which may compromise the credibility of their role as evaluators.[2] Furthermore, we conduct experiments for human preferences by crowdsourcing six human annotators and collecting each of their rankings for a total of 300 annotations. From

[1]Our project page: https://github.com/minnesotanlp/cobbler

[2]In total, **42K** samples are analyzed across six biases for each model totaling **630K** samples.

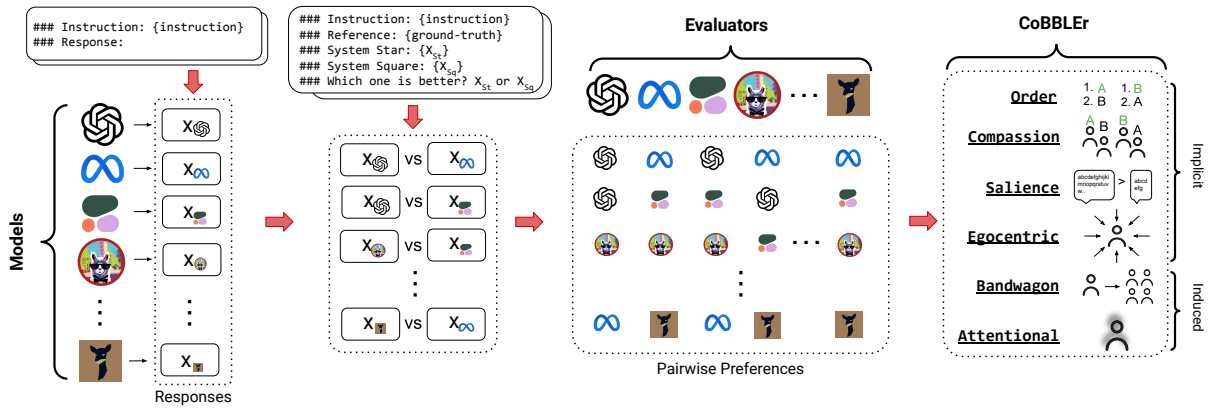

Figure 1: Our CoBBLEr pipeline to evaluate popular LLMs that are instruction-tuned or trained with human feedback for their capabilities as unbiased automatic evaluators.

our findings, we observe a low correlation between human and machine judgments via Rank-Biased Overlap (RBO), indicating that machine and human preferences are generally in low agreement.

Our core contributions are as follows:

- A new benchmark (CoBBLEr) for evaluating LLMs to perform unbiased evaluations within the QA setting.
- An examination of an exhaustive list of 6 (cognitive) evaluation biases that have not been covered by previous studies. We find that most LLMs cannot perform as unbiased evaluators.
- A comprehensive lineup of models (sizing from $3B$ to $>175B$ parameters) as evaluators, encompassing the current state-of-the-art language models covering over **630k** comparisons.

From our benchmark, we find that most models exhibit various cognitive biases when used as automatic evaluators, which may negatively impact evaluation quality. Thus, we propose our benchmark (CoBBLEr) for measuring the capabilities of language models as evaluators to enable more reliable evaluations that are well-aligned with human judgment.

We note that our use of *biased* and *unbiased* preferences does not allude to the ability to make completely impartial judgments but rather the *amplification* of human-like biases within language models. As most models are tuned on human data, our study aims to estimate this gap between model and human judgment such that they can be refined more effectively to mitigate against these biases. As such, we also aim for our benchmark to be applied towards the development of future models, as in discovering new gaps or finding that existing gaps are still unresolved.

2 Related Work

LLMs as Evaluators. Owing to the effectiveness of LLMs, many recent research works have investigated their utility in various downstream tasks, such as machine translation (Kocmi and Federmann, 2023), summarization (Shen et al., 2023; Gao et al., 2023), code generation (Zhuo, 2023), writing assistance (Schick et al., 2023; Raheja et al., 2023), factual consistency (Cohen et al., 2023; Gekhman et al., 2023; Luo et al., 2023), and more. Additionally, many studies have leveraged LLMs for general-purpose NLG evaluation. For instance, Liu et al. (2023); Chen et al. (2023); Wang et al. (2023a) investigated the effectiveness of GPT-4 and ChatGPT against reference-free evaluation methods, whereas Fu et al. (2023) proposed an evaluation framework, GPTScore, to score generated texts. Recently, Li et al. (2023a) and Zheng et al. (2023) conducted similar experiments by employing LLMs as evaluators to judge the quality of generations in a pairwise setting. Although these works present promising results for LLMs as automatic evaluators, our work takes a closer look at machine artifacts that could be detrimental to data quality by benchmarking an exhaustive list of biases impacting LLMs-as-evaluators.

LLM Evaluation Benchmarks. It is becoming increasingly challenging to evaluate open-source LLMs as they become more powerful and performant. As a result, there has been an increasing need to develop better evaluation benchmarks for measuring the performance of LLMs. However, most of these benchmarks, such as LM-Eval-Harness (Gao et al., 2021), MMLU (Hendrycks et al., 2021), HELM (Liang et al., 2022) and BIG-

Bias	Bias Behavior	Example
ORDER BIAS	The tendency to give preference to an option based on their order (e.g. first, second, or last).	**System Star**: x **System Square**: y System Square: y System Star: x
COMPASSION FADE	The tendency to observe different behaviors when given recognizable names as opposed to anonymized aliases.	Model Alpaca: x Model Vicuna: y **Model Vicuna**: y **Model Alpaca**: x
EGOCENTRIC BIAS	The inclination to prioritize one's own responses regardless of response quality.	**Model Star (You)**: x Model Square: y
SALIENCE BIAS	The tendency to prefer responses based on the length of the response (i.e., more often preferring longer responses over shorter ones).	**System Star**: The quick brown fox jumps over the lazy dog. System Square: The fox jumped.
BANDWAGON EFFECT	The tendency to prefer majority belief without critical evaluation.	**85%** believe that System Star is better.
ATTENTIONAL BIAS	The inclination to give more attention to irrelevant or unimportant details.	System Square **likes to eat oranges and apples**

Table 1: We display the characteristic format for each bias and bold answers that indicate behavior influenced by the bias. For example, in COMPASSION FADE (recognizable names) Model Alpaca and Model Vicuna are associated with System Star and System Square respectively, in which the preferred response (bolded) is inconsistent with the preferred response from ORDER (anonymized names). Specific prompt details are viewed in Appendix C.

BENCH (Srivastava et al., 2023), only focus on general LLM performance but do not explore their capabilities as evaluators. Our work in this direction overlaps directly with Bai et al. (2023) and Zheng et al. (2023), who propose a Language-Model-as-an-Examiner benchmark and LLM-as-a-judge to study the capability of LLMs to emulate human preferences. While our experimental setups are similar, we highlight key differences. We cover a wider demographic of current popular language models and an overall different focus on QA as opposed to other domains such as math and reason. Furthermore, our benchmark emphasizes a wider range of biases (implicit/induced) to better describe machine artifacts when used as automatic evaluators. Specifically, COBBLER measures the extent to which each LM-as-evaluator is impacted in each decision by certain artifacts within prompts (i.e., prompting format, prompt information) over a comprehensive list of cognitive biases.

Cognitive Biases in LLMs. While biases have been well-known to exist in LLMs (Wang et al., 2023b; Talboy and Fuller, 2023; Wu and Aji, 2023), many recent works investigating the behaviors of LLMs have also uncovered similarities with cognitive biases. Some recent works (Zhao et al., 2021; Liu et al., 2022; Lu et al., 2022) have shown that the order of training examples in GPT-3 could lead to differences in accuracy between near chance and near state-of-the-art. Jones and Steinhardt (2022) captured failures in GPT-3 and Codex and found that error patterns of LLMs resemble cognitive biases in humans. Our work overlaps with these in some of the biases we cover, but we present a much more holistic and comprehensive evaluation of LLMs. Along this aspect, while our work is close to Wu and Aji (2023), who investigate biases related to fabricated factual and grammatical errors, our work is much more comprehensive in terms of the number of LLMs analyzed, the types of biases analyzed and the creation of an open benchmark.

3 COBBLER: Cognitive Bias Benchmark for LLMs as Evaluators

The following criteria are used to select each type of evaluation bias:

- **General Applicability.** Text evaluation tasks should be generalizable to most prompting scenarios; tasks that observe too specific subtleties within the prompt are not helpful.
- **Impartiality.** The prompt should not involve any leading statements to extract some desired quality of the evaluations
- **Memorylessness.** The current evaluation instance should not rely on any previous behaviors. Each instance should be self-contained when extracting each bias metric.

We carefully hand-select these biases based on the above three criteria such that they can be widely applicable to most evaluation settings in assessing the performance of LLMs as automatic evaluators.